

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL ¹ - Concept Mapping

Course Code:	CSA0311	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
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Section / Batch:	2025	Date:	13/07/26

Code of Conduct
I, M. C. Dhineshkumar (192524424) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: M. C. Dhineshkumar

Instructions

- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

[Signature]
13/7/26

1).

SMART - HEALTHCARE SYSTEM

Data Objects (real-entities/ attributes).
Patient ID, Name, Age, Gender, Temp BP
Diagnosis.

Data structure (way to store Data)

Primitive data structure (single value data type) int, float, char, bool, double

ABSTRACT DATA TYPE (ADT)
(Define data + operation) stack, queue, list.

Non Primitive data structure (collection values) Array, Linked List, Trees.

Stack
(LIFO - Last In First Out)
→ used for undo operation

Operations:-
Push()
Pop()
Peek()
isEmpty()

Operation
Enqueue()
Dequeue()
Front()
isEmpty()

Queue
(FIFO - First In First Out)
→ Used for patient flow management.

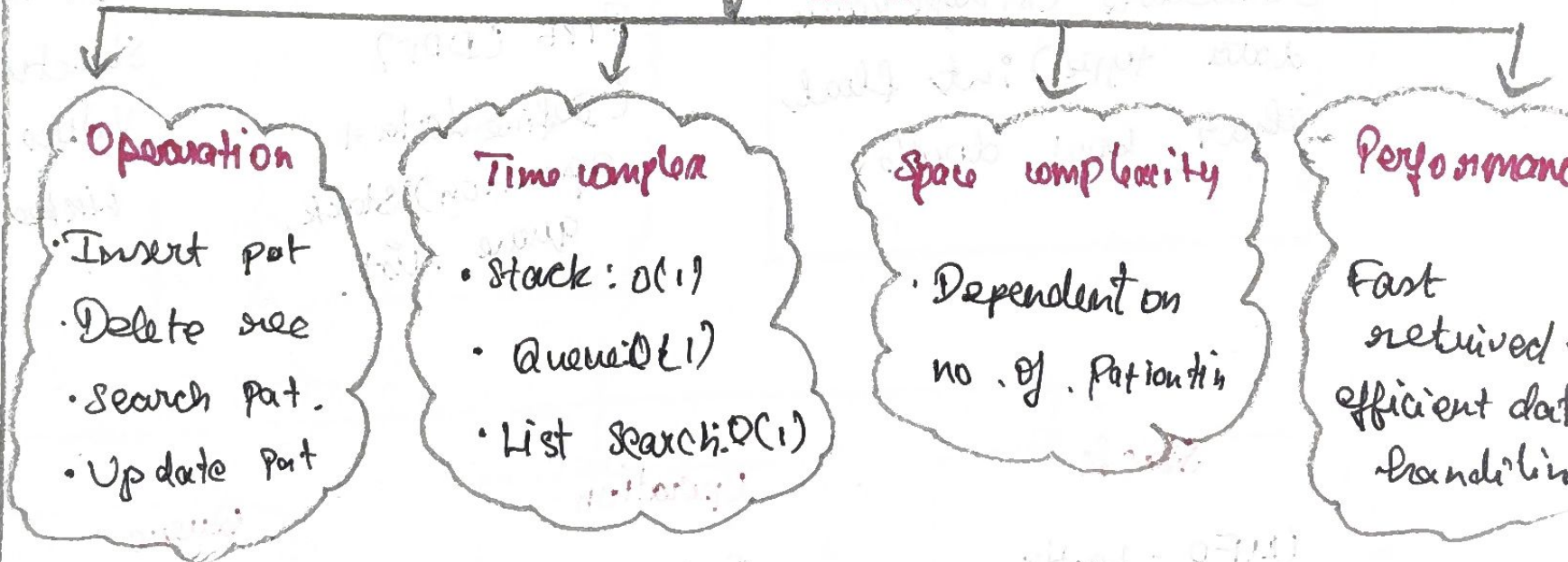
Operation
Insert()
Delete()
Search()
Traverse()

List.
(linear collection) → used for storing records

Implementation In C - language.

Structure	Pointer	Function	Dynamic memory
<pre> struct patient{ int id; char name[50]; int age; float temp; } </pre>	used to link nodes in linked list <pre> list {Tree struct node next; } </pre>	• Insert() • Delete() • Search() • Update() • Display()	• malloc • calloc • free

Leads to.



SMART CITY TRAFFIC MANAGEMENT.

Data Structure
(The way of organizing data)

Primitive Data Structure
(Basic type of data)
int, float, char, double, bool

Non-Primitive Data Structure
(Made up of collection of value of arrays, linked list, Stack, Queue, tree)

NON-PRIMITIVE DATA STRUCTURE

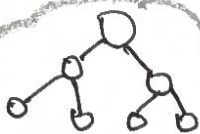
Linear Data Structure
Element arrange in seq

Non Linear Data Structure
(Elements in hierarchy)

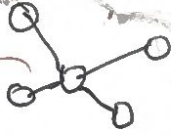
ARRAY


-> Fixed size of similar element
eg: count[100];

TREE


-> Hierarchical structure.
eg: Road hierarchy

GRAPH.


-> Network of nodes connect by edges.

Linked List

→ Dynamic collection of elements.
eg:- Store list of vehicles detect by sensors.

Queue



(FIFO)

Used for waiting lines.

eg:- Vehicles wait at traffic signals.

APPLICATIONS IN SMART CITY TRAFFIC

Vehicle Monitoring

• Store and process vehicle data from sensor.

Traffic Signal Control

• Queue data helps signal scheduling

ROUTE OPTIMIZATION

Graph help in find shortest / fastest

CONGESTION DETECTION

Analyse traffic density using and graphs.

Handwritten signature and date: 13/11/24